TASK 2 – Choose the Data Types

2.1 Data Type Selection

1. Previous meter reading: unsigned short, because it stores non-negative meter readings.
2. Current meter reading: unsigned short, because it stores readings from 0 to 65535.
3. Units consumed: unsigned int, because it stores the calculated electricity consumption.
4. Household number: unsigned int, because it identifies a household.
5. Consumer category: char, because it can store a category code.
6. Energy charge: long long, because it stores monetary values in paise.
7. Fixed charge: long long, because it stores the fixed charge in paise.
8. Total bill: long long, because it stores the sum of all charges in paise.

2.2 Type Sizes and Ranges

Typical GCC sizes are char = 1 byte, short = 2 bytes, int = 4 bytes, long long = 8 bytes, float = 4 bytes, and double = 8 bytes. Actual sizes depend on the system. The limits.h and float.h header files provide information about type ranges and floating-point properties. The sizeof operator is used to determine the size of a data type.

2.3 Why Money Should Not Be Stored as Float

Floating-point numbers cannot represent every decimal amount exactly. Small errors may accumulate when thousands of bills are added. Storing money as integer paise avoids decimal rounding errors, provided the integer type does not overflow.

2.4 Meter Rollover

A 16-bit unsigned meter register has values from 0 to 65535. After 65535, it rolls over to 0. Unsigned arithmetic wraps around modulo 65536.

For a previous reading of 65400 and a current reading of 120:

Consumption = 65536 − 65400 + 120 = 256 units.

Therefore, the consumption is 256 units and not a negative number.

2.5 Alternative Concept and Complexity Analysis

Adding double values takes O(1) time per addition but may introduce rounding errors. Integer-paise addition is exact when overflow does not occur. Both types normally use a fixed amount of memory. Summing n bills takes O(n) time. Rounding should be performed at a clearly defined stage in the billing process.

Conclusion

Choosing suitable data types helps prevent incorrect meter calculations and improves billing accuracy. Unsigned short is suitable for a 16-bit meter register, while long long integer values in paise help avoid floating-point errors in monetary calculations.

CODE:

```
#include <stdio.h>
#include <limits.h>

typedef unsigned short meter_t;

unsigned int consumption(meter_t previous, meter_t current)
{
   return (meter_t)(current - previous);
}

int main(void)
```

```
{
    printf("Size of meter_t: %zu bytes\n", sizeof(meter_t));
    printf("Maximum reading: %u\n", (unsigned int)USHRT_MAX);

    printf("Consumption: %u units\n", consumption(1200, 1530));
    printf("Consumption: %u units\n", consumption(65400, 120));

    return 0;
}
```

